

NYC Taxi Lakehouse Platform - Master Blueprint

Project Vision

The goal of this project is to build an **end-to-end enterprise-grade Data Engineering Lakehouse platform** using the NYC Taxi public dataset. The platform will simulate real-world Data Engineering practices by implementing:

- Data ingestion
- Incremental processing
- Data quality validation
- Delta Lake capabilities
- Performance optimization
- Monitoring and logging
- Business analytics

The solution will follow the **Medallion Architecture** (Bronze, Silver, Gold) and demonstrate production-ready patterns commonly used in Databricks and modern data platforms.

Architecture Overview

Landing Zone

The project begins by ingesting raw NYC Taxi datasets into a **Landing Zone** where files are preserved exactly as received.

Bronze Layer

A **Bronze Layer** is created using Delta tables to store raw data along with ingestion metadata such as: - Load timestamps - Source file information

Silver Layer

The **Silver Layer** contains cleansed, validated, and transformed datasets. Data engineering operations include: - Deduplication - Standardization - Schema enforcement - Type casting - Derived column generation

The data will be partitioned appropriately to optimize storage and query performance.

Gold Layer

The **Gold Layer** serves as the business and analytics layer. Aggregated datasets will be created to answer business questions related to: - Revenue trends - Trip patterns - Demand forecasting - Peak travel periods - Customer behavior - Geographic analysis - Operational efficiency

Data Loading Strategies

The platform will support both **Full Load** and **Incremental Load** strategies:

Strategy	Description
Full Load	Used during the initial ingestion of all historical data
Incremental Load	Subsequent executions use watermark-based incremental loading with a metadata table tracking previously processed records

This approach reflects industry-standard ETL and ELT design patterns to reduce processing costs and improve scalability.

Data Quality Framework

A reusable **Data Quality Framework** validates incoming data before promotion to downstream layers:

Check	Description
Null validation	Ensure critical columns are not null
Duplicate detection	Identify and remove duplicate records
Datatype verification	Confirm column types match expected schema
Schema validation	Enforce structural consistency
Business rule enforcement	Validate domain-specific constraints
Negative value checks	Flag invalid negative values in numeric fields
Record count reconciliation	Cross-check row counts between layers
Audit logging	Persist all quality results to audit tables

Results are stored in quality audit tables for observability and reporting.

Delta Lake Capabilities

Capability	Purpose
ACID transactions	Reliable concurrent reads and writes
Schema enforcement & evolution	Prevent bad data; allow controlled schema changes
MERGE operations	Upsert (insert + update) patterns
Time Travel	Query historical table versions
Optimize	Compact small files for faster reads
Vacuum	Remove old unused files to reclaim storage
Transaction history inspection	Audit and debug changes via Delta log
Version management	Rollback or compare data across versions

Performance Optimization

- Partitioning by relevant columns (e.g., date)
- Caching frequently accessed DataFrames
- Broadcast joins for small dimension tables
- Query plan analysis via `explain()`
- Adaptive Query Execution (AQE) concepts
- File compaction strategies
- Delta Lake `OPTIMIZE` and `ZORDER BY` commands

Production Engineering Practices

Practice	Description
Pipeline execution logs	Track every run with status and timing
Monitoring tables	Centralized metrics for observability
Audit trails	Record all data movements and transformations
Metadata-driven processing	Config-driven pipeline for extensibility
Error handling mechanisms	Graceful failure with retry and alerting

Reporting & Analytics

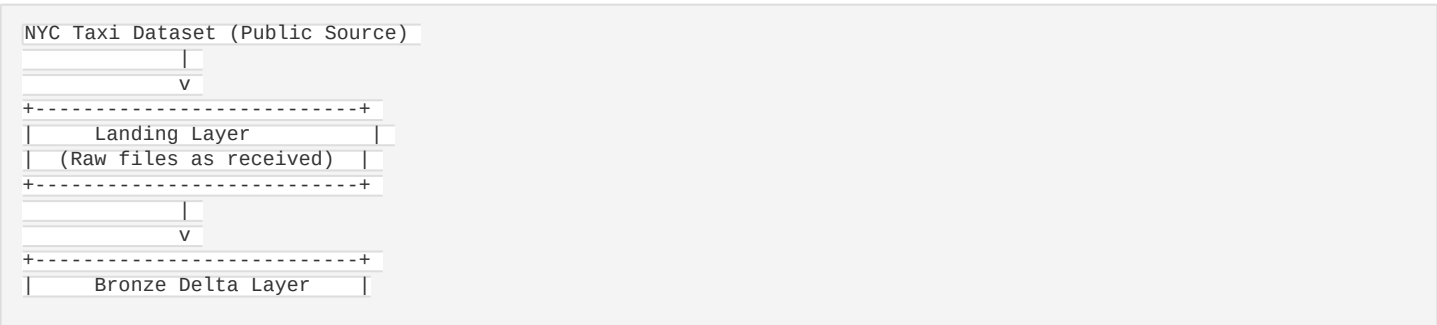
The Gold Layer datasets feed a **reporting and analytics dashboard** built using Power BI or another free visualization tool. The dashboard provides executive-level insights and showcases the business value generated from the platform.

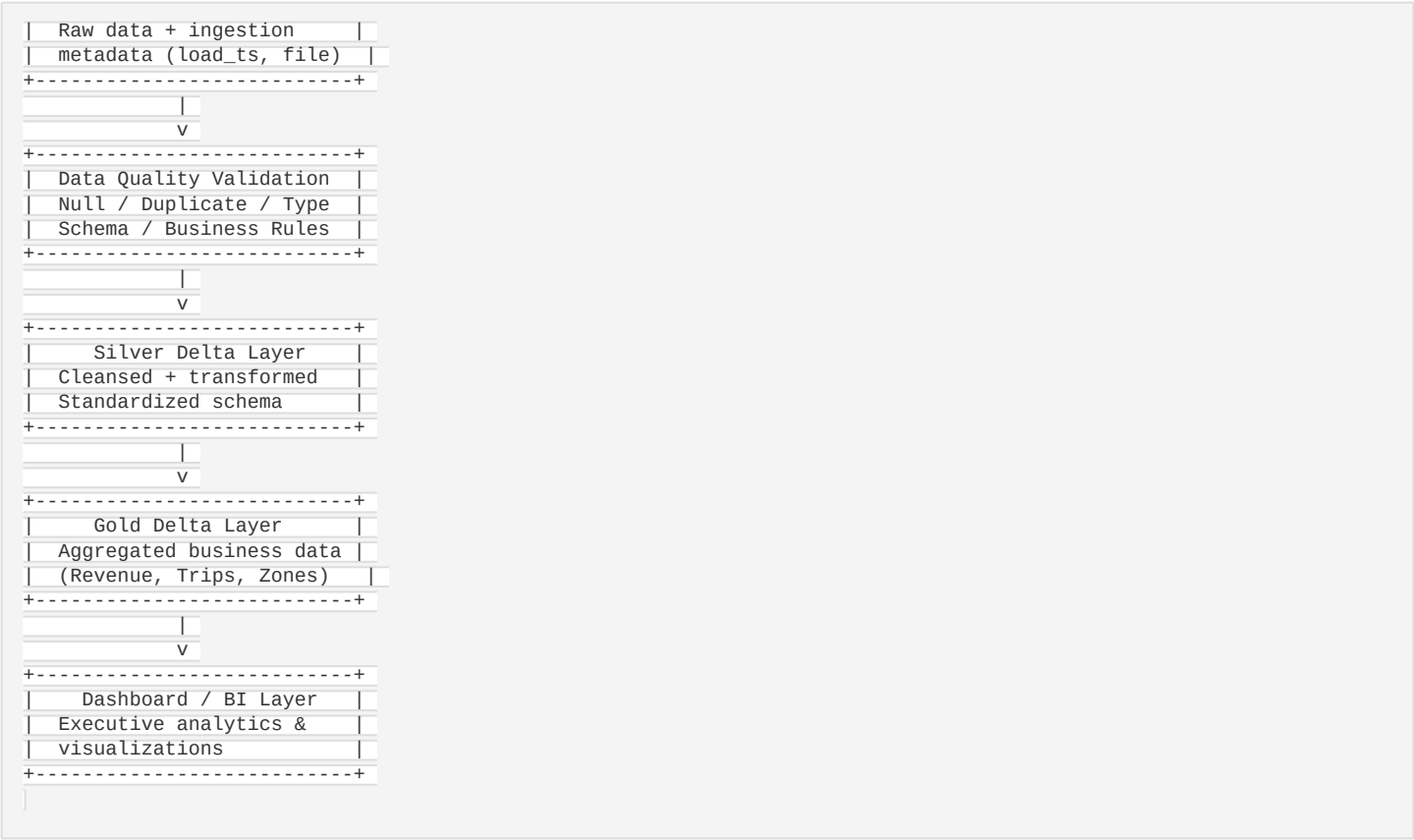
Skills Demonstrated

Category	Skills
Core Processing	PySpark, Spark SQL, Delta Lake
Architecture	Medallion Architecture, ETL/ELT Design, Data Platform Design
Data Engineering	Incremental Processing, Data Modeling, Metadata-Driven Pipelines
Quality & Observability	Data Quality Framework, Data Observability, Audit Logging
Performance	Partitioning, Broadcast Joins, Optimization, Caching
Analytics & BI	Business Intelligence, Dashboarding, Reporting
Tooling	Databricks Concepts, Monitoring, Error Tracking

Architecture Flow & Phase-wise Roadmap

Data Pipeline Flow





Phase-wise Implementation Roadmap

Phase	Title	Key Deliverables
1	Project Setup	Download NYC Taxi dataset, create project structure, configure PySpark + Delta Lake environment
2	Landing & Bronze Layer	Create landing zone, implement full load, store raw Delta tables, add ingestion metadata
3	Incremental Load Framework	Metadata table, watermark strategy, incremental processing, pipeline tracking
4	Data Quality Framework	Null checks, duplicate checks, datatype validation, business rule validation, audit tables
5	Silver Layer	Data cleansing, transformations, standardized schema, partitioning
6	Gold Layer	Daily revenue, monthly revenue, peak hours, zone analysis, trip analytics
7	Delta Lake Features	Merge, time travel, optimize, vacuum, schema evolution
8	Performance Optimization	Broadcast joins, caching, repartitioning, explain plans
9	Monitoring & Logging	Pipeline logs, metadata tables, error tracking, audit reporting

Phase	Title	Key Deliverables
10	Dashboard	Revenue analytics, trip analytics, operational analytics, executive dashboard

Interview Value Mapping

This project demonstrates hands-on proficiency across **19 key areas**:

#	Skill	#	Skill
1	PySpark	11	Partitioning
2	Spark SQL	12	Broadcast Join
3	ETL	13	Optimization
4	ELT	14	Data Pipeline Design
5	Data Modeling	15	Audit Logging
6	Incremental Loads	16	Monitoring
7	Metadata-Driven Processing	17	Business Intelligence
8	Delta Lake	18	Dashboarding
9	Medallion Architecture	19	Data Platform Architecture
10	Data Quality Framework		